Mechanical Prototyping

Activity 2

Case Study of RPT

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Title:-

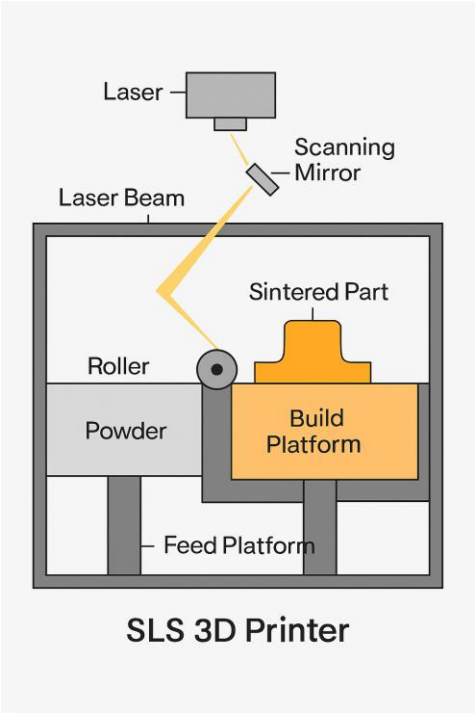
Case Study: SLS for Rapid Prototyping of **BMW** Engine Water Pump Housing

1. Real Example:-

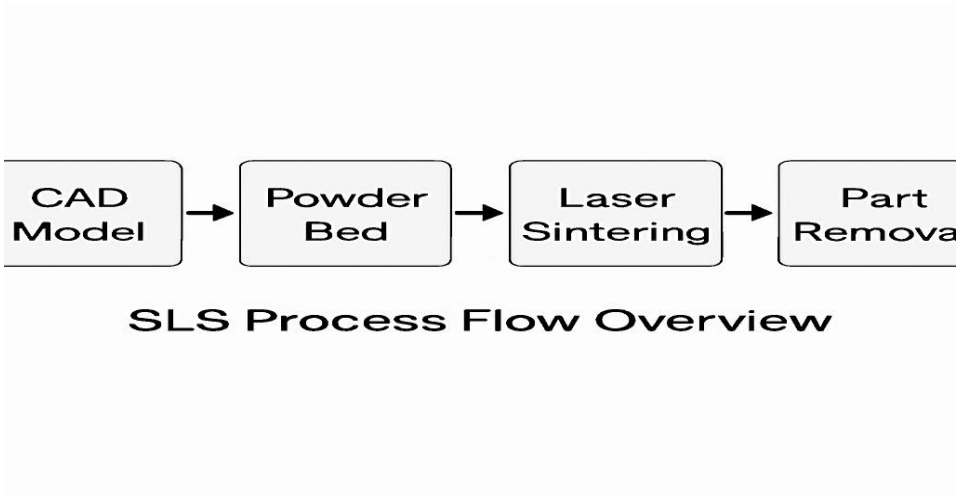
• In **2018**, **BMW's** powertrain team used **SLS 3D printing** to prototype **water pump housings** in just days—slashing development time by **60%**.

2. Technique / Method of Rapid Prototyping:-

Selective Laser Sintering (SLS) — an additive process where a high-power laser fuses **nylon powder (PA12)** layer by layer, creating strong, heat-resistant parts **without support structures**.



SLS 3D Printer



SLS Process Flow Overview

3. Aim of Case Study :-

To explore how **BMW** leveraged **SLS** to move from **concept to functional engine component** in days, enabling **real-world testing** of coolant flow and thermal performance before investing in **metal tooling**.


SLS Water Pump Housing

4. Why SLS Was Used in the Industry :-

- **Material performance: PA12** delivers mechanical strength and thermal stability comparable to injection-molded parts, making prototypes suitable for **engine-bench testing**.

- **Geometric freedom:** Complex internal **coolant passages** and undercuts print cleanly in a **powder bed**, eliminating the need for removable supports.
- **Rapid iteration:** Engineers received printed housings within **48 hours**, versus weeks for **CNC** or cast molds, accelerating design refinement.
- **Cost savings:** Early identification of design flaws prevented expensive **metal tooling changes**, saving an estimated **$50 000** in upfront expenses.

5. Details of Case Study :-

Industry Context & Challenge:

BMW's engine division faced a tight schedule to validate a redesigned **water pump housing**. Traditional **CNC machining** and low-volume casting each took **3–4 weeks** per iteration and required custom fixtures. Every delay cascaded into the wider **vehicle launch** timeline.

SLS Implementation & Workflow:

1. **Collaborative design sprint:**
   A cross-functional team (designers, **CFD analysts**, test engineers) sketched three housing concepts focused on **optimized coolant flow** and reduced pressure drop.

2. **CAD to print:**
   Engineers exported each design as an **STL file** and queued them on an **EOS P 396 SLS machine** using **PA12 powder**. Overnight printing produced all three variants simultaneously.

3. **Post-processing:**
   Technicians extracted un-sintered powder and **bead-blasted** surfaces to achieve smooth inlet/outlet faces. No support removal was needed, preserving delicate internal channels.

4. **Functional testing:**
   • **Engine-bench rig:** Prototypes fitted to a test engine to record **flow rate**, **pressure endurance**, and **temperature cycles** up to **120 °C**.
   • **Dimensional verification:** A **3D scan** confirmed critical tolerances of **± 0.2 mm** against the CAD model.

5. **Iterative feedback:**
   Within **two weeks**, the team identified a stress-concentration area near the mounting flange. They refined the geometry, printed a second iteration, and validated the fix—all before ordering a single metal tool.

Project Team Perspective:
Project lead **Martina Fischer** recalls, “For the first time, we saw physical water pump housings in our hands faster than our production planners expected drawings.” Test engineers noted that the nylon prototypes behaved so similarly to molded parts that they trusted the results—boosting confidence and team morale.

Comparative Analysis

Method	Lead Time per Iteration	Material Waste	Functional Fidellty
SLS	2–3 days	<10% powder	Engine-bench ready
CNC	3–4 weeks	~40% stock	Ra 3–5 µm
Casting	4–6 weeks	Mould cost	Ra 5–10 µm
	4–6 weeks	Mould cost	Tool-dependent

6. Conclusion:-

BMW's adoption of **SLS** turned rapid prototyping into "**functional validation on demand**." By printing engine-ready parts in days, they reduced financial risk, accelerated product development, and empowered engineers to iterate boldly. This case underscores SLS's strategic value in **high-performance industries** where **real-world testing** and **speed** are paramount.